Week 2: Comparing Groups

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2026-06-25

Skills Learning – Lecture

Last week, we focused on identifying different types of variables and thinking about how the structure of our data affects the type of analysis we can use.

This week, we will focus on one major branch of the statistical decision tree:

> **Comparing groups when the response variable is quantitative.**

Today we will focus on:

- one-sample t-tests
- independent two-sample t-tests
- paired t-tests
- ANOVA
- visualizing group comparisons
- interpreting results biologically

The main question for today is:

> **I have a quantitative response variable and I want to compare groups. Which test should I use?**

0. Load Required Packages

```r
library(tidyverse)
library(janitor)
library(here)
```

1. Load and Preview the Dataset

Today we will continue using the faux gorilla hormone + behavioral dataset.

Each row represents one sample or observation from a gorilla. Some gorillas appear more than once, which means the dataset contains repeated measures. Later in the course, we will learn how to model repeated measures directly. Today, we will focus on the logic of comparing groups.

```r
# load dataset
data <- readr::read_csv(here::here("Week 1/gorilla_hormone_data.csv")) %>%
  janitor::clean_names()
```

```
## Rows: 240 Columns: 19
## -- Column specification ------------------------------------
## Delimiter: ","
## chr  (8): sample_id, gorilla_id, gorilla_name, group, sa...
## dbl (11): age_years, dominance_rank, group_size, fruit_a...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# preview dataset
glimpse(data, width = 80)
```

```
## Rows: 240
## Columns: 19
## $ sample_id               <chr> "G001_S01", "G001_S02", "G001_S03", "G001_S04~
## $ gorilla_id              <chr> "G001", "G001", "G001", "G001", "G001", "G001~
## $ gorilla_name            <chr> "Gasore", "Gasore", "Gasore", "Gasore", "Gaso~
## $ group                   <chr> "BWE", "BWE", "BWE", "BWE", "BWE", "BWE", "BW~
## $ sample_date             <chr> "5/23/25", "8/18/25", "9/10/25", "2/2/25", "5~
## $ observer                <chr> "Mark", "Sarah", "Mary", "Sarah", "Mark", "Ro~
## $ sex                     <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age_class               <chr> "adult", "adult", "adult", "adult", "adult", ~
## $ age_years               <dbl> 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.~
## $ dominance_rank          <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 6, 6, 6, 6, 6, 6, 6, ~
## $ group_size              <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, 3, ~
## $ fruit_availability_index <dbl> 0.4040977, 0.3301009, 0.3134359, 0.7098761, 0~
## $ feeding_time_prop       <dbl> 0.2554745, 0.6272515, 0.5106832, 0.3779914, 0~
## $ aggression_events       <dbl> 2, 0, 1, 0, 0, 1, 1, 0, 3, 0, 1, 0, 0, 1, 0, ~
## $ parasite_load           <dbl> 28, 3, 12, 19, 15, 7, 38, 26, 6, 17, 10, 4, 5~
## $ respiratory_signs       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ injury_status           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
## $ cortisol_ng_ml          <dbl> 19.158612, 22.606257, 28.816896, 9.639142, 10~
## $ testosterone_ng_ml      <dbl> 5.717721, 3.666248, 3.084256, 4.783930, 3.218~
```

2. What Does "Comparing Groups" Mean?

A group comparison asks whether a quantitative response variable differs across categories.

For example:

- Do injured and uninjured gorillas differ in cortisol?
- Do males and females differ in testosterone?
- Do adults, subadults, and juveniles differ in cortisol?

The **response variable** is the numeric variable we are comparing.

The **grouping variable** is the categorical variable that defines the groups. ### 2.1 summary stats

```r
# Examples of quantitative response variables
summary(data$cortisol_ng_ml)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   6.985  12.570  16.370  18.050  21.590  55.701
```

```r
summary(data$testosterone_ng_ml)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.526   3.788   5.855   7.152   9.084  34.719
```

```r
# Examples of grouping variables
data %>% count(injury_status)
```

```
## # A tibble: 2 x 2
##   injury_status     n
##           <dbl> <int>
## 1             0   220
## 2             1    20
```

```r
data %>% count(sex)
```

```
## # A tibble: 2 x 2
##   sex       n
##   <chr> <int>
## 1 F       136
## 2 M       104
```

```r
data %>% count(age_class)
```

```
## # A tibble: 3 x 2
##   age_class     n
##   <chr>     <int>
## 1 adult       216
## 2 infant        8
## 3 subadult     16
```

3. One-Sample t-test

A one-sample t-test compares the mean of one group to a known or expected value.

Example question:

> Is the average cortisol value different from 20 ng/mL?

Here, we only have one sample of cortisol values, and we are comparing the mean to a known value: 20.

Why are we comparing it to 20 ng/mL?
For a real biological interpretation, the comparison value would need to come from somewhere meaningful, such as:

- a published population average
- a known physiological threshold
- a historical baseline from the same population
- a management or conservation benchmark

For example, if previous research found that healthy adult mountain gorillas have an average cortisol concentration of 20 ng/mL; then we want to test whether our population is different than that observed mean.

3.1 Test

```r
# One-sample t-test
t.test(data$cortisol_ng_ml, mu = 20)
```

```
##
##  One Sample t-test
##
## data:  data$cortisol_ng_ml
## t = -3.8372, df = 239, p-value = 0.0001593
## alternative hypothesis: true mean is not equal to 20
## 95 percent confidence interval:
##  17.04824 19.05087
## sample estimates:
## mean of x
##  18.04956
```

This test asks:

> Is the mean cortisol value in our sample statistically different from 20 ng/mL?

Important pieces of output:

- **mean of x**: the observed sample mean
- **confidence interval**: plausible range for the true mean
- **p-value**: evidence against the idea that the true mean equals 20

The `mean` cortisol value in our dataset is: **18.05 ng/mL**, which is lower than **20 ng/mL**.

The test then asks: *Is this difference large enough that it is unlikely to be due to random sampling variation alone?*

The `p-val` = **0.00016**, which is « 0.05.
This suggests that the mean cortisol level in our sample is likely different from 20 ng/mL.

The `confidence interval` is: **17.05 to 19.05 ng/mL**, which is our best estimate of where the true population mean probably lies.
Notice that 20 ng/mL is *not* inside this interval. That is another clue that the true mean is probably different from 20 ng/mL.

> Biological interpretation: The average cortisol concentration in our sample was approximately 18.1 ng/mL. This was significantly lower than the reference value of 20 ng/mL (one-sample t-test, $p < 0.001$). The estimated population mean cortisol concentration is likely between 17.0 and 19.1 ng/mL.

The most important question is not "Is the p-value significant?" The most important question is: What is the estimated mean, and how different is it from the value we are comparing it to?

We can then say: *The gorillas in our sample had a mean cortisol concentration of 18.1 ng/mL, which was significantly lower than the published reference value of 20 ng/mL ($p < 0.001$).*

or

The average cortisol concentration in our sample was significantly below this threshold, suggesting that the population was not experiencing unusually elevated stress during the sampling period.

4. Visualizing Group Comparisons

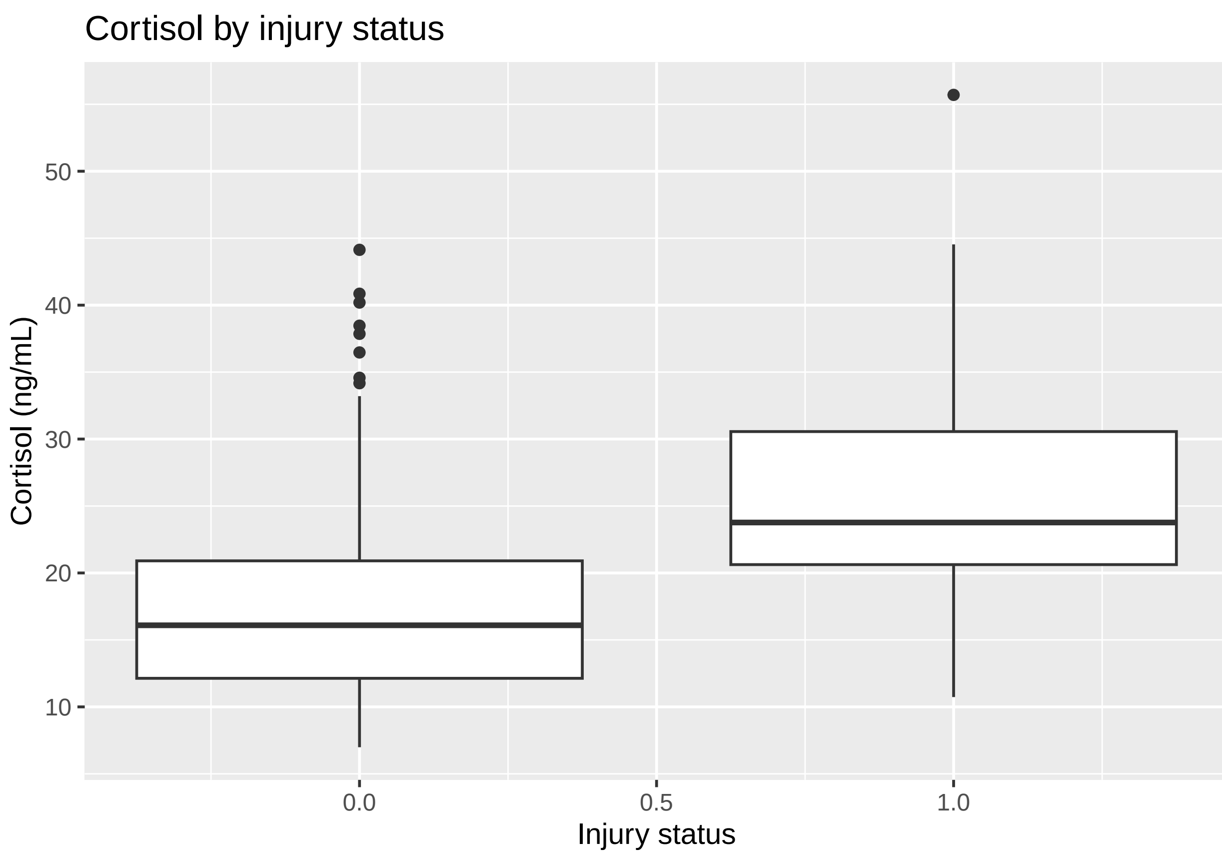
Before running any statistical test, we should always visualize the data.

Common plots for comparing groups include:

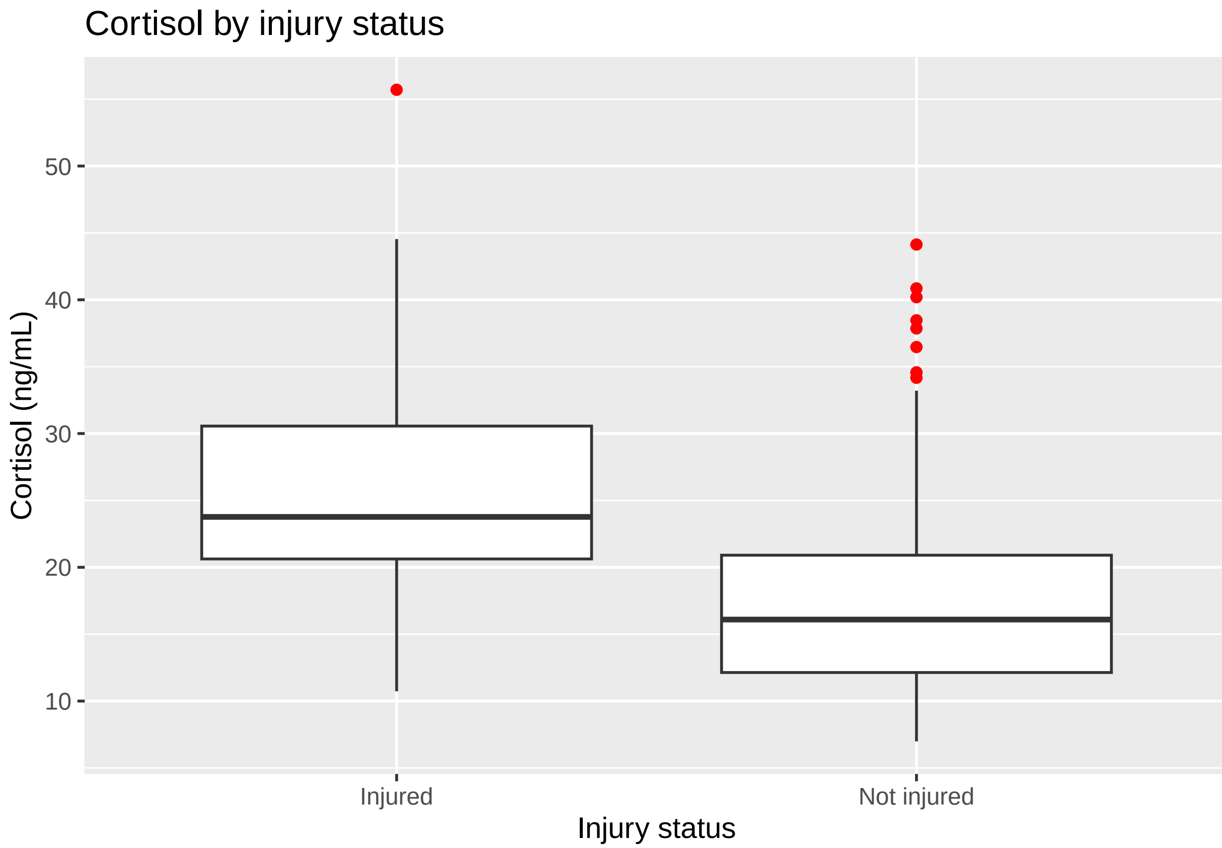
- boxplots
- jitter plots

4.1 Boxplot

```r
ggplot(data, aes(x = injury_status, y = cortisol_ng_ml, group = injury_status)) +
  geom_boxplot() +
  labs(
    x = "Injury status",
    y = "Cortisol (ng/mL)",
    title = "Cortisol by injury status"
  )
```




```r
# Make injury labels easier to read
data <- data %>%
  mutate(
    injury_label = if_else(injury_status == 1, "Injured", "Not injured")
  )

ggplot(data, aes(x = injury_label, y = cortisol_ng_ml)) +
  geom_boxplot(outlier.colour = "red") +
  labs(
    x = "Injury status",
    y = "Cortisol (ng/mL)",
    title = "Cortisol by injury status"
  )
```



5. Independent Two-Sample t-test

An independent two-sample t-test compares the mean of a quantitative variable between two independent groups.

Example question:

> Do injured gorillas have higher cortisol than uninjured gorillas?

Response variable:

- `cortisol_ng_ml`

Grouping variable:

- `injury_label`

Number of groups:

- 2

Design:

- independent groups

5.1 Independent t-test

```
# Independent two-sample t-test
t.test(cortisol_ng_ml ~ injury_label, data = data)
```

```
##
##  Welch Two Sample t-test
##
## data:  cortisol_ng_ml by injury_label
## t = 3.8853, df = 20.4, p-value = 0.0008925
## alternative hypothesis: true difference in means between group Injured and group Not injured is not e
## 95 percent confidence interval:
##   4.545507 15.056200
## sample estimates:
##     mean in group Injured mean in group Not injured
##                  27.03367                  17.23282
```

This test asks:

> Is the average cortisol level different between injured and uninjured gorillas?

The p-value tells us whether the difference between group means is larger than we would expect from random variation alone.

The confidence interval gives a plausible range for the difference between group means.

Look at the group means:

- Mean cortisol (Injured) = 27.03 ng/mL
- Mean cortisol (Not injured) = 17.23 ng/mL

Before even looking at the p-value: *injured gorillas* appear to have **higher cortisol** levels than *uninjured gorillas*. The difference is: 27.03 - 17.23 = **9.80 ng/mL**

> So injured gorillas have, on average, about 10 ng/mL higher cortisol.

The t-test asks: *is this difference large enough that it is unlikely to be due to random sampling variation alone?*

- p-value: We see that **p « 0.05**, so, there is strong evidence that cortisol differs between injured and uninjured gorillas.
- 95% confidence interval: **4.55 - 15.06**. So, We estimate that injured gorillas have cortisol values somewhere between 4.5 and 15.1 ng/mL.
- notice that '0' is not inside the CI; so, this is a clue that the groups differ. If 0 was within the CI, then statistically, there was a possibility that the difference between the groups was zero!

We can say that *injured gorillas have significantly higher cortisol concentrations than uninjured gorillas (Welch's two-sample t-test, p < 0.001). On average, injured gorillas have cortisol concentrations approximately 9.8 ng/mL higher than uninjured gorillas. The estimated difference between groups is between 4.5 and 15.1 ng/mL.*

Important to note: **Does this mean injury causes elevated cortisol?**
Not necessarily!

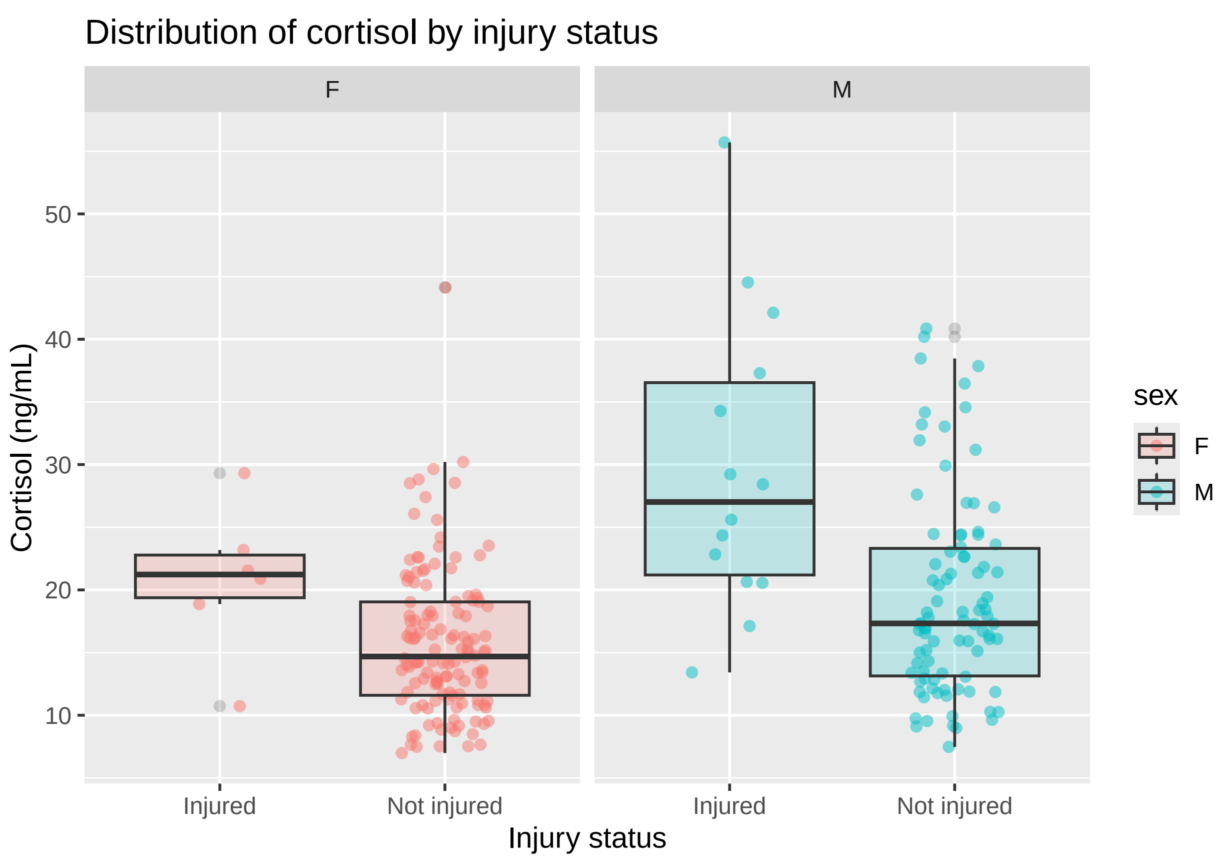
Possible explanations:

- injury increases physiological *stress*, which elevates cortisol
- individuals with higher cortisol may be more likely to become injured
- another factor may influence *both* injury risk and cortisol

  Remember: association is not the same thing as causation.

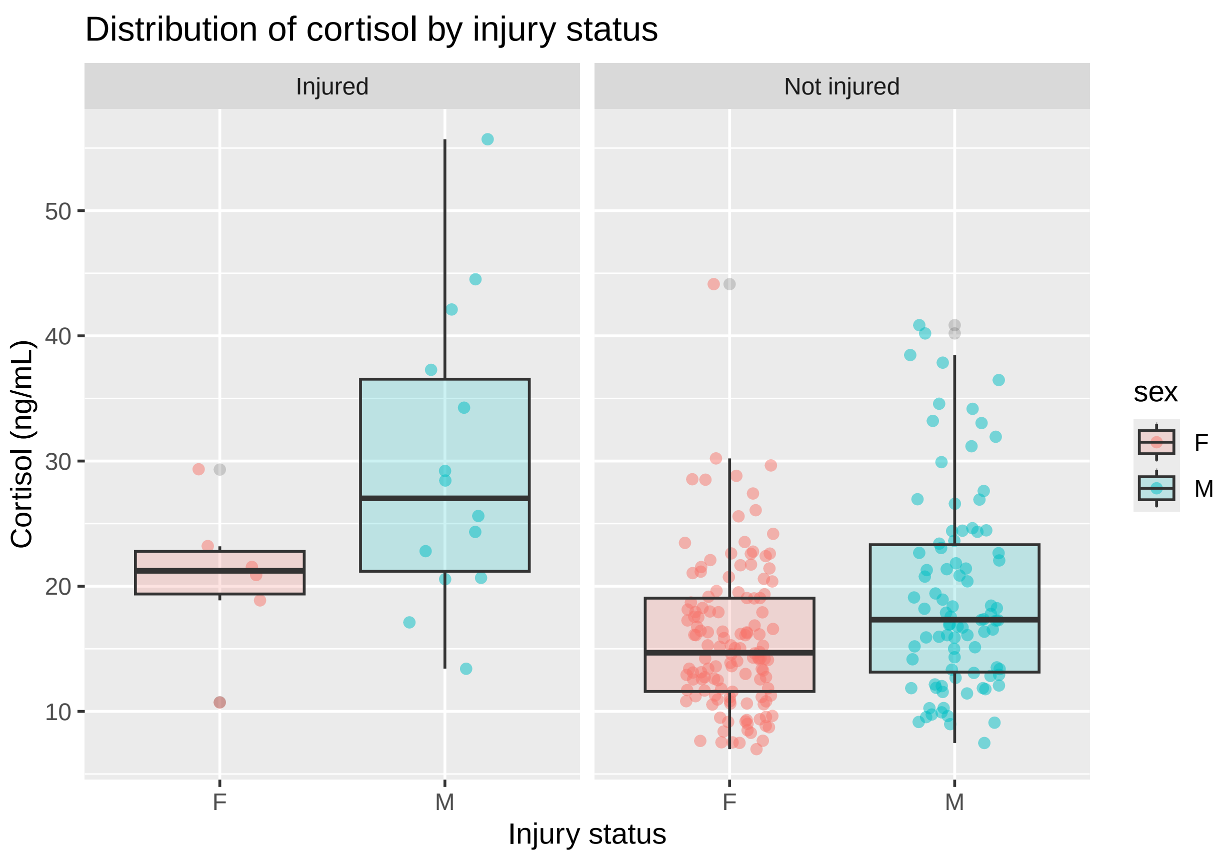
5.2 Plots

```
# can also be a boxplot
ggplot(data, aes(x = injury_label, y = cortisol_ng_ml)) +
  geom_jitter(alpha = 0.5, width = 0.2, aes(color = sex)) +
  geom_boxplot(alpha = 0.2, aes(fill = sex)) +
  facet_wrap(~sex) +
  labs(
    x = "Injury status",
    y = "Cortisol (ng/mL)",
    title = "Distribution of cortisol by injury status"
  )
```




```
# you can also change the perspective by faceting by injury status instead of sex.
ggplot(data, aes(x = sex, y = cortisol_ng_ml)) +
  geom_jitter(alpha = 0.5, width = 0.2, aes(color = sex)) +
  geom_boxplot(alpha = 0.2, aes(fill = sex)) +
  facet_wrap(~injury_label) +
```

```r
  labs(
    x = "Injury status",
    y = "Cortisol (ng/mL)",
    title = "Distribution of cortisol by injury status"
  )
```



6. Paired t-test

A paired t-test is used when each observation in one group has a natural partner in the other group. For example:

> What if we measured cortisol in the same gorillas before and after an injury?

In that case, each gorilla has two measurements:

- before injury
- after injury

Those values are paired because they come from the same individual.

6.1 Import paired data

```r
paired_data <- read_csv(here::here("Week 2/gorilla_paired.csv")) %>%
  janitor::clean_names()
```

```
## New names:
## Rows: 15 Columns: 4
## -- Column specification
## ------------------------------------ Delimiter: "," chr
## (1): gorilla_id dbl (3): ...1, before_injury, after_injury
## i Use `spec()` to retrieve the full column specification
## for this data. i Specify the column types or set
## `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

6.2 Run t-test

```r
# Paired t-test
t.test(
  paired_data$after_injury,
  paired_data$before_injury,
  paired = TRUE
)
```

```
##
##  Paired t-test
##
## data:  paired_data$after_injury and paired_data$before_injury
## t = 14.616, df = 14, p-value = 7.168e-10
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  4.436952 5.963048
## sample estimates:
## mean difference
##             5.2
```

Ask yourself:

> Can I draw a line connecting one value to another because they belong to the same individual?

- If yes, the data may be paired.
- If no, the groups are probably independent.

Our question is: *Do cortisol levels change after an injury?* Importantly, we are **not** comparing two different groups of gorillas. We're comparing the same individuals before and after an event.

The paired t-test therefore focuses on the difference within each individual.

> Mean difference = 5.2

Biological interpretation: *on average, cortisol increased by about 5.2 ng/mL after injury.*

Notice that the test is not reporting the mean before or the mean after. Instead, it calculates: (after - before) for each gorilla and then analyzes those differences.

- p-value: « **0.05**, so we can say that *cortisol levels appear to increase after injury.*

- 95% confidence interval: **4.44 to 5.96**, so the true average increase in cortisol is likely somewhere between: 4.44 and 5.96 ng/mL.

- notice that '0' is not inside the CI; so, this is a clue that cortisol appears to change with injury. If injury had no effect, we would expect the mean difference to be close to zero.

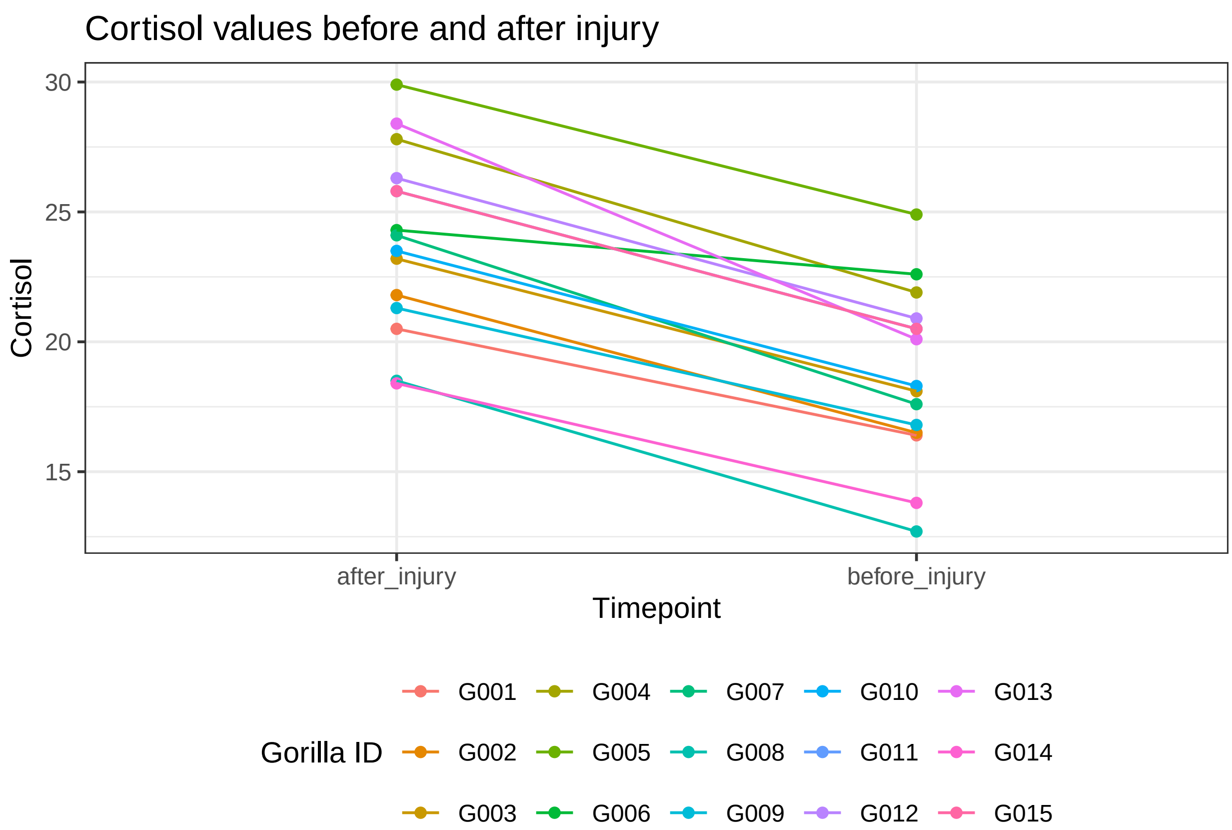
To summarize: *Cortisol concentrations were significantly higher after injury than before injury (paired t-test, p < 0.001). On average, cortisol increased by approximately 5.2 ng/mL following injury. The estimated increase was between 4.44 and 5.96 ng/mL.*

6.3 Plot

```r
# Reshape for plotting
paired_long <- paired_data %>%
  pivot_longer(
    cols = c(before_injury, after_injury),
    names_to = "timepoint",
    values_to = "cortisol"
  )

paired_long
```

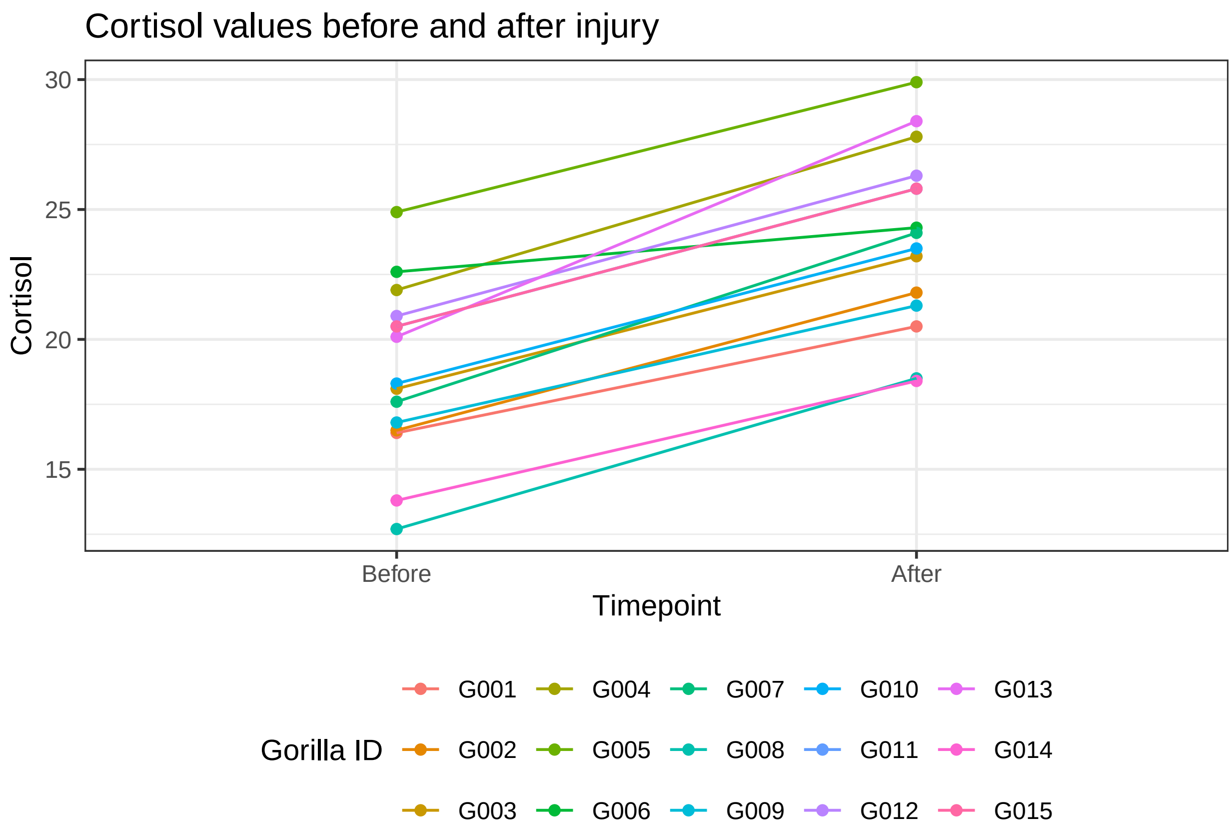
```
## # A tibble: 30 x 4
##       x1 gorilla_id timepoint     cortisol
##    <dbl> <chr>      <chr>            <dbl>
##  1     1 G001       before_injury     16.4
##  2     1 G001       after_injury      20.5
##  3     2 G002       before_injury     16.5
##  4     2 G002       after_injury      21.8
##  5     3 G003       before_injury     18.1
##  6     3 G003       after_injury      23.2
##  7     4 G004       before_injury     21.9
##  8     4 G004       after_injury      27.8
##  9     5 G005       before_injury     24.9
## 10     5 G005       after_injury      29.9
## # i 20 more rows
```

```r
# plot as paired data on x-axis
ggplot(paired_long, aes(x = timepoint, y = cortisol, group = gorilla_id, color = gorilla_id)) +
  geom_point() +
  geom_line() +
  labs(
    x = "Timepoint",
    y = "Cortisol",
    title = "Cortisol values before and after injury",
    color = "Gorilla ID"
  ) +
  theme_bw() +
  theme(legend.position = "bottom")
```




```
# but we see that the order of the labels on the x-axis isn't correct,
#chronologically (we usually read left-to-right)...

# so, let's change that: We'll change `timepoint` (the x variable) to a
# factor with a particular order:

# Before, `timepoint` was a character. A character variable is simply text.
# R treats each value as a word or label. No particular meaning or order.

# A factor variable is a categorical variable. It still looks like text
# to us, but now R knows that the values represent categories that can be compared.

paired_long <- paired_long %>%
  mutate(timepoint = factor(timepoint, levels = c("before_injury", "after_injury")))

# now let's plot again
ggplot(paired_long, aes(x = timepoint, y = cortisol, group = gorilla_id, color = gorilla_id)) +
  geom_point() +
  geom_line() +
  labs(
    x = "Timepoint",
    y = "Cortisol",
    title = "Cortisol values before and after injury",
    color = "Gorilla ID"
  ) +
  scale_x_discrete(labels = c("Before", "After")) +
```

```r
  theme_bw() +
  theme(legend.position = "bottom")
```



7. ANOVA: Comparing 3 or More Groups

ANOVA is used when we want to compare the mean of a quantitative response variable across three or more groups.

Example question:

> Do cortisol levels differ among age classes?

Response variable:

- `cortisol_ng_ml`

Grouping variable:

- `age_class`

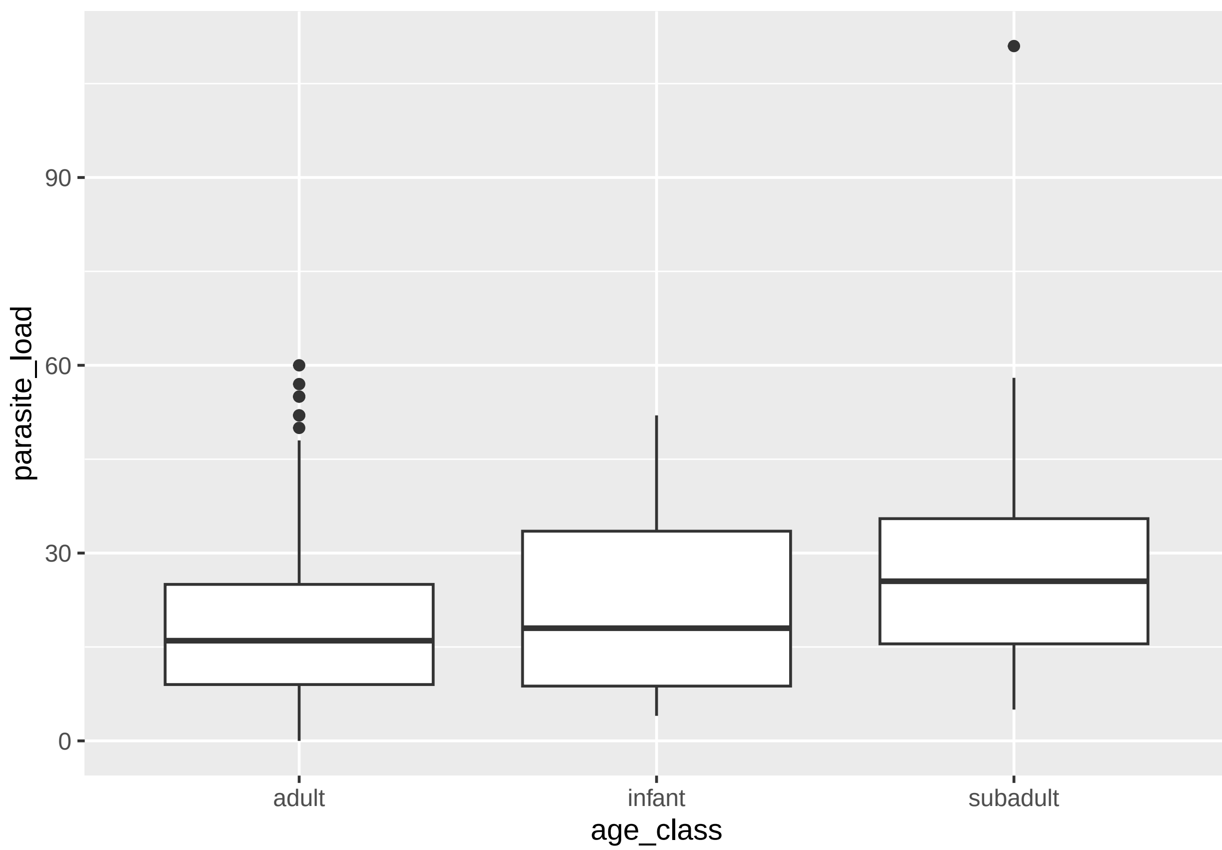
Number of groups:

- 3 or more ### 7.1 Plot

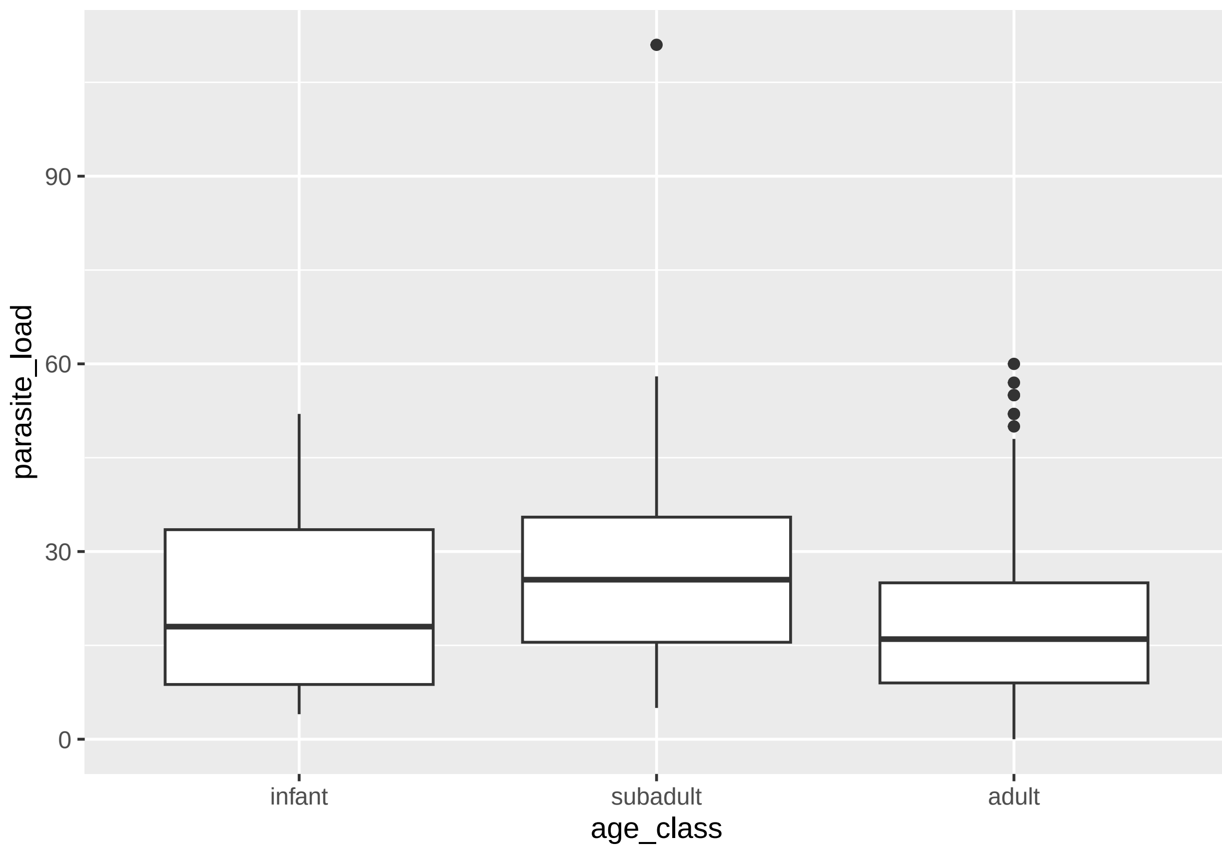
```
data %>%
  count(age_class)
```

```
## # A tibble: 3 x 2
##   age_class     n
##   <chr>     <int>
## 1 adult       216
## 2 infant        8
## 3 subadult     16
```

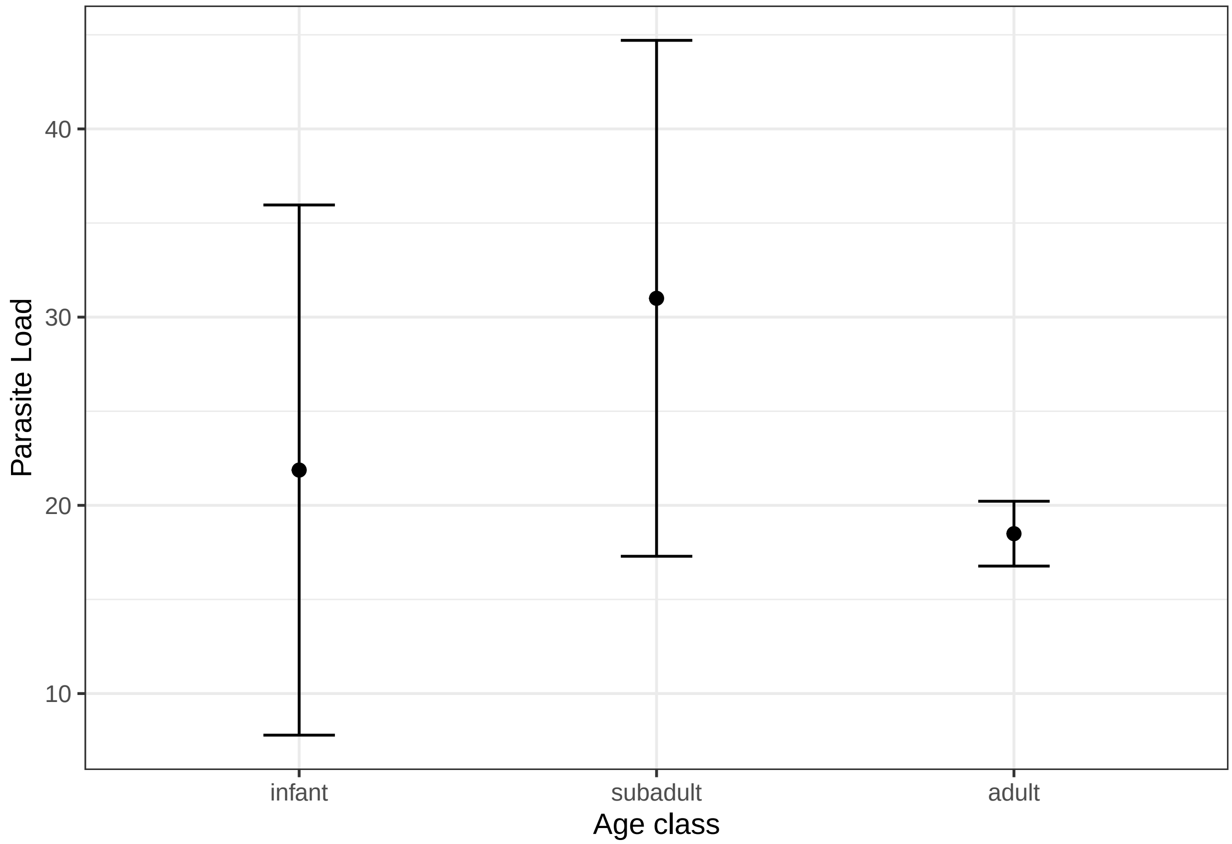
```
ggplot(data, aes(x = age_class, y = parasite_load)) +
  geom_boxplot()
```




```
data <- data %>%
  mutate(age_class = factor(age_class, levels = c("infant", "subadult", "adult")))

# yes, we can look at boxplots: the *distribution* of the data
ggplot(data, aes(x = age_class, y = parasite_load)) +
  geom_boxplot()
```




```r
# but a better, statistical representation of the means (which we are comparing), is a stat_summary:
ggplot(data, aes(age_class, parasite_load)) +
  stat_summary(fun = mean,
               geom = "point",
               size = 2
               ) +
  stat_summary(fun.data = mean_cl_normal,
               geom = "errorbar",
               width = 0.2
               ) +
  labs(
    x = "Age class",
    y = "Parasite Load",
  ) +
  theme_bw()
```



Boxplots: Maximum (within 1.5 x IQR)
|
|
Q3 — <- top of box (75th percentile)
|
Median - <- middle line (50th percentile)
|
Q1 — <- bottom of box (25th percentile)
|
|
Minimum (within 1.5 x IQR)

Whereas `geom_boxplot()` shows distribution, `stat_summary(fun = mean)` ignores quartiles and just looks at the actual average + error (uncertainty due to variation in the data).

- Boxplot: *"What do the data look like?"*
- Mean +/- CI: *"How precisely have we estimated the population mean?"*

7.2 ANOVA

```
# Fit ANOVA model
mod_aov <- aov(parasite_load ~ age_class, data = data)

# View ANOVA table
summary(mod_aov)
```

```
##              Df Sum Sq Mean Sq F value  Pr(>F)
## age_class     2   2378  1189.2   5.951 0.00301 **
## Residuals   237  47361   199.8
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA asks:

> Are any of the group means different from each other?

ANOVA does not compare every pair of groups individually. It only asks: *looking at all three groups together, is there evidence that at least one mean differs from the others?*

It does **not** tell us *which* groups are different. For that, we need a follow-up comparison. It first asks whether there is evidence of any difference among the means.

- p-values: **Pr(>F) = 0.003**. This is *much* smaller than 0.05, so we have strong evidence that parasite load differs among age classes.

However, ANOVA does not tell us *which* age classes differ. It only tells us that **at least one difference exists** somewhere among the groups.

To determine which specific groups differ, we need a post-hoc test:

It asks: *if the ANOVA found differences among the groups, which specific groups differ?*

So Tukey performs all pairwise comparisons:

- Infant vs Adult
- Subadult vs Adult
- Subadult vs Infant

…while correcting for the fact that we are making multiple comparisons. Without that correction, we'd increase our chance of false positives.

Because Tukey performs multiple statistical tests, it adjusts the p-values to reduce the chance of finding a significant result simply by chance.

Think of it this way:

- If we perform one statistical test, there is about a 5% chance of finding a significant result by chance alone (when using a = 0.05).
- If we perform many tests, those chances begin to add up. The more comparisons we make, the greater the chance of getting a false positive.

Tukey corrects for this by adjusting the p-values so that the *overall* probability of making even one false positive across *all* pairwise comparisons remains close to 5%.

Instead of treating each comparison independently, **Tukey treats them as one family of comparisons**, which is why the output states: 95% *family-wise* confidence level

7.3 Tukey

HSD = Honestly Significant Difference

```r
# Pairwise comparisons after ANOVA
TukeyHSD(mod_aov)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = parasite_load ~ age_class, data = data)
##
## $age_class
##                        diff       lwr       upr     p adj
## subadult-infant    9.12500  -5.31205 23.562050 0.2971922
## adult-infant      -3.37963 -15.38374  8.624480 0.7845709
## adult-subadult   -12.50463 -21.14306 -3.866197 0.0021654
```

- The `diff` column gives the estimated difference between the two group means.
- The `lwr` and `upr` columns give the 95% confidence interval for that difference.
- The `p adj` column contains the *adjusted* p-value, which accounts for the fact that multiple pairwise comparisons are being made.

Infant vs. Adult

- Difference = 3.38 parasites
- Adjusted p = 0.785

There is no evidence that parasite loads differ between adults and infants.

Subadult vs. Adult

- Difference = 12.50 parasites
- Adjusted p = 0.002

Subadults had, on average, about 12.5 more parasites than adults, and this difference was statistically significant.

Notice that the 95% confidence interval (3.87 to 21.14) does *not* include zero, providing further evidence that the groups differ.

Subadult vs. Infant

- Difference = 9.13 parasites
- Adjusted p = 0.297

Although subadults had a higher average parasite load than infants, the difference was not statistically significant.

Overall interpretation:

1. The ANOVA showed that parasite load differed among age classes (p = 0.003).
2. The Tukey HSD test showed that this overall difference was driven by subadults having significantly higher parasite loads than adults.
3. No significant differences were detected between adults and infants or between subadults and infants after correcting for multiple comparisons.

   You should only run a Tukey test if you first have a reason to ask which groups differ; i.e., if the ANOVA came back significant.

8. Checking Assumptions

For t-tests and ANOVA, we often want to know whether the response variable is approximately normally distributed within groups.

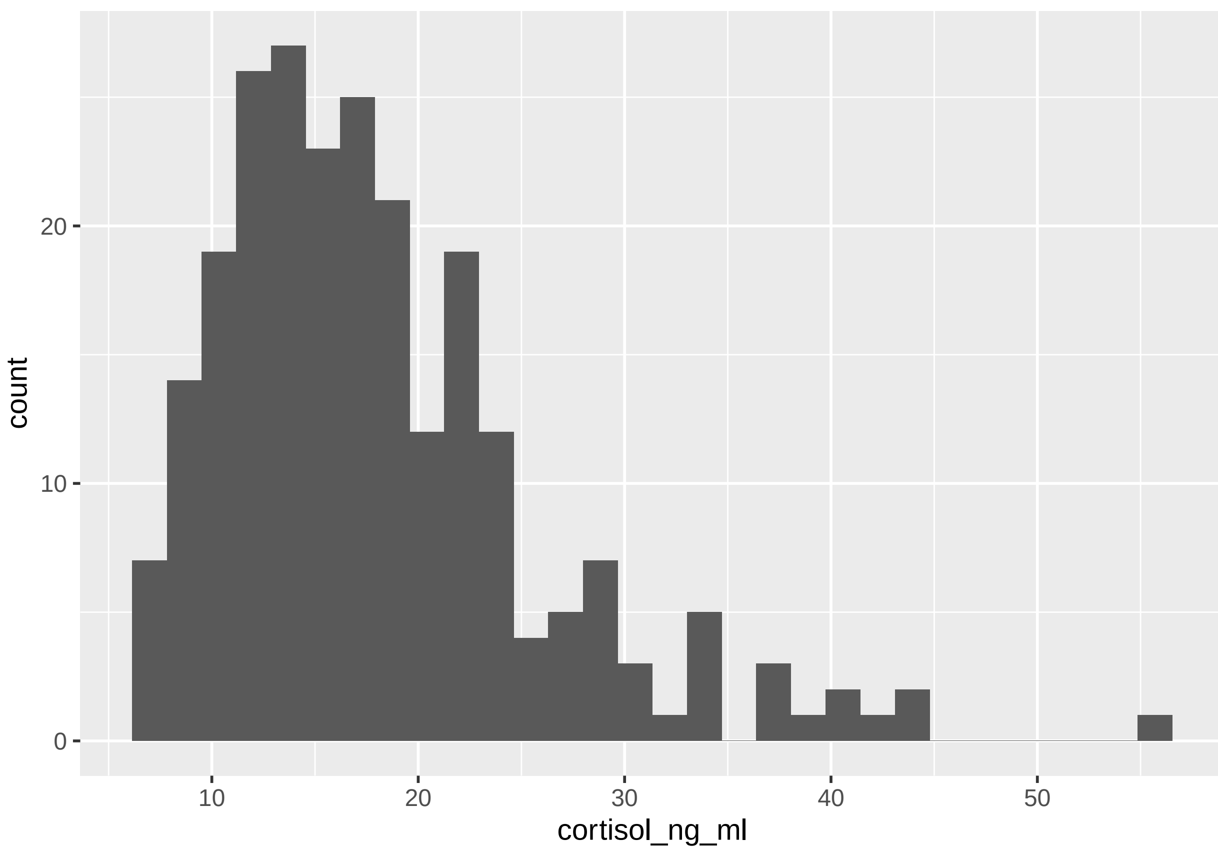
We also want to know whether the spread of the data is reasonably similar across groups.

Today, we will keep this simple.

8.1 Histograms

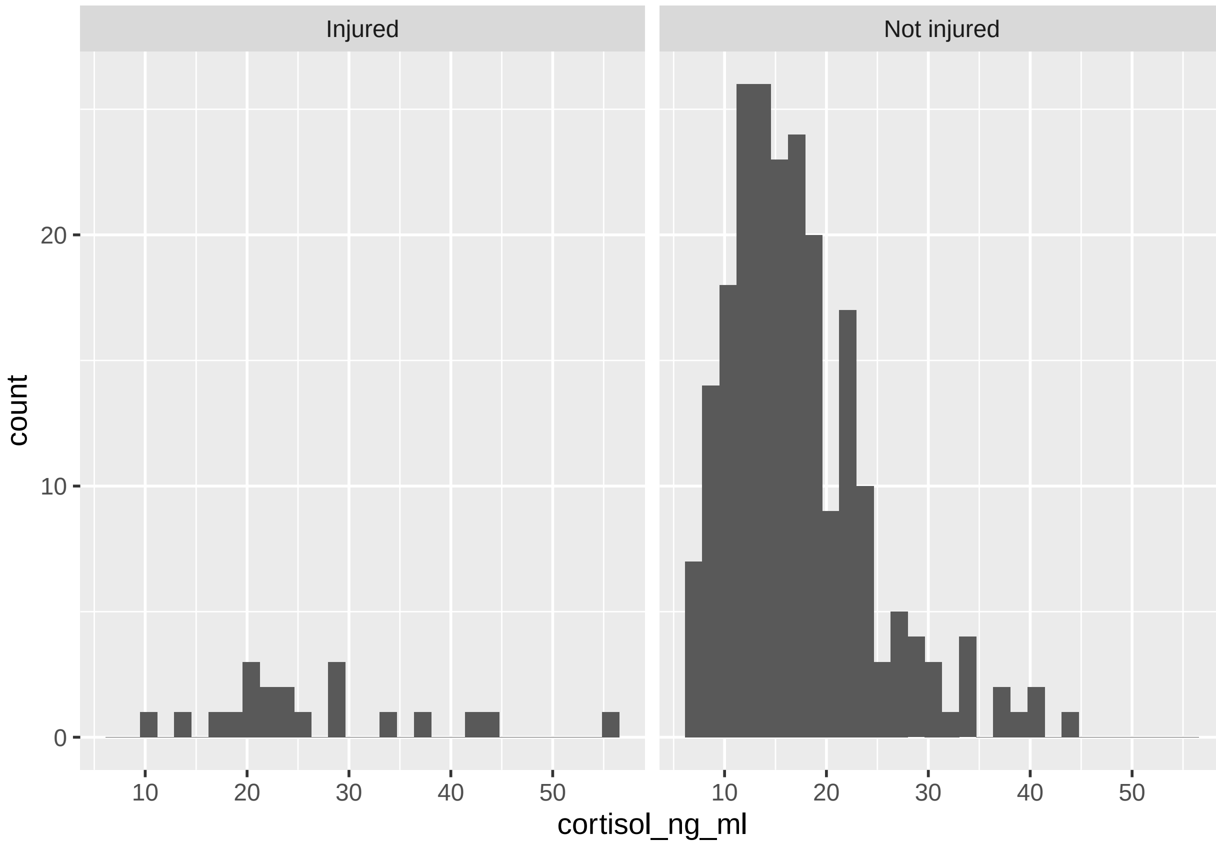
```
ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value
## `binwidth`.
```




```
ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_histogram() +
  facet_wrap(~ injury_label)
```

```
## `stat_bin()` using `bins = 30`. Pick better value
## `binwidth`.
```



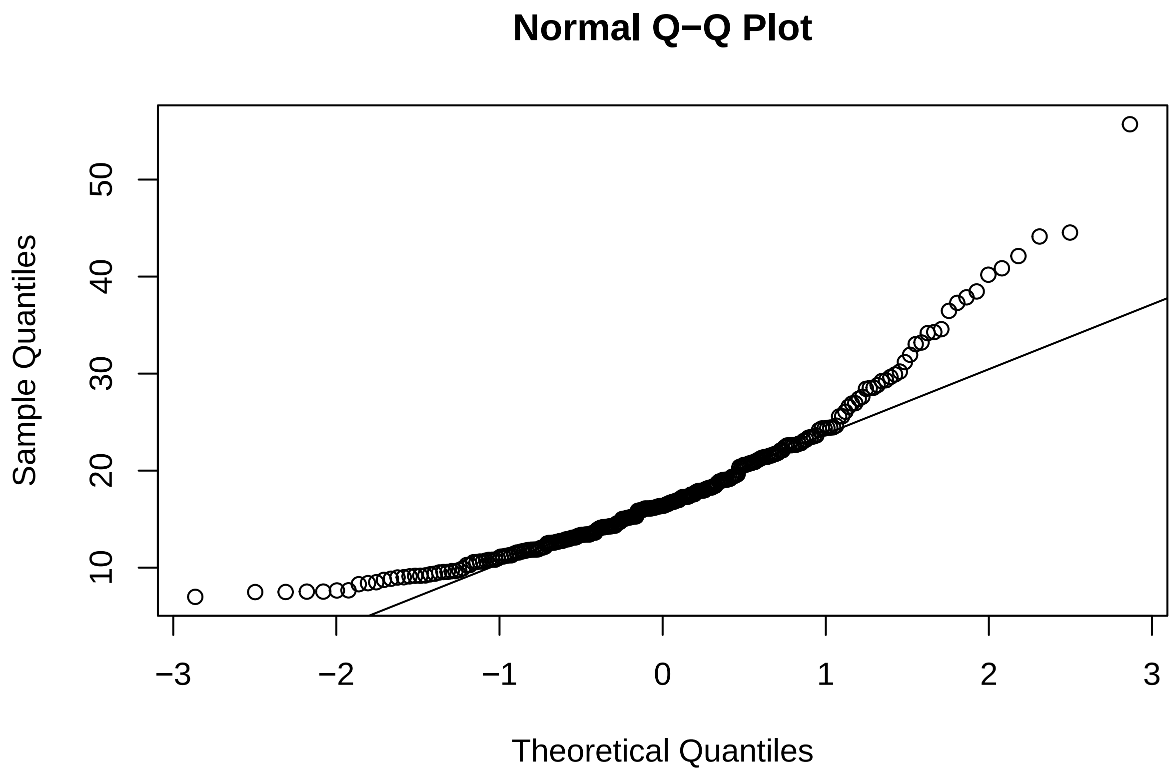
8.2 QQ Plot

A QQ plot helps us check whether the data are approximately normal. It compares:

- what our data look like
- what perfectly normal data would look like

If the data were perfectly normal, the points would fall roughly along the straight line.

```
qqnorm(data$cortisol_ng_ml)
qqline(data$cortisol_ng_ml)
```



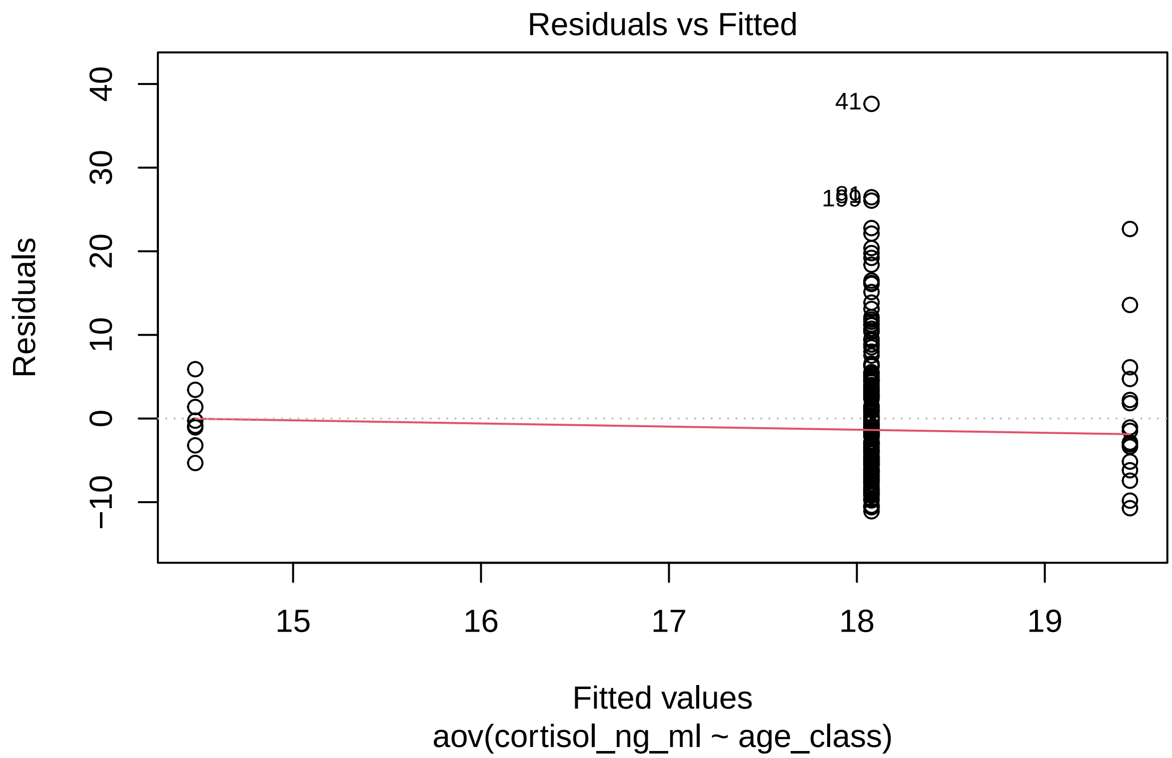
QQ plot: What we see is that the points follow the line fairly well in the middle curve upward at the upper end and have one large point at the very top

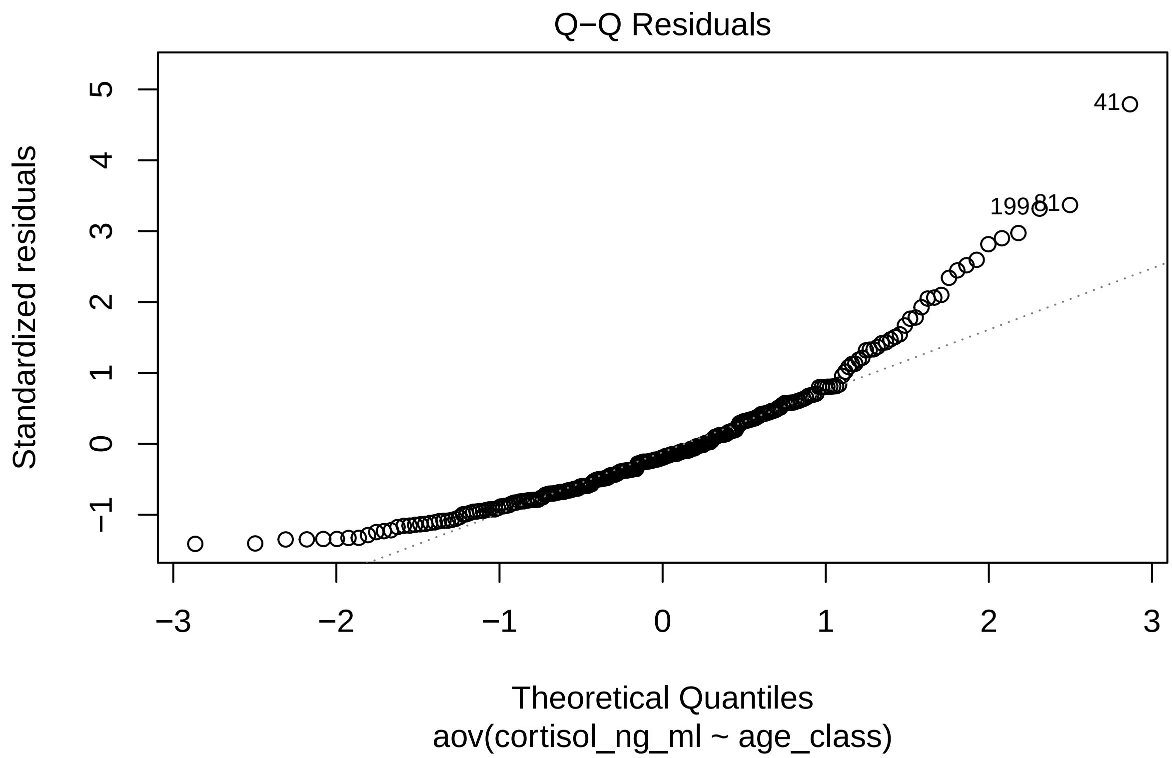
This tells us: *The data are approximately normal through the middle of the distribution, but have a heavier upper tail (some unusually large values).*

We can see these unusually large values in the histogram! The data are not perfectly normal, but they are *reasonably close*. There are a few unusually high observations that pull the upper tail away from the line.

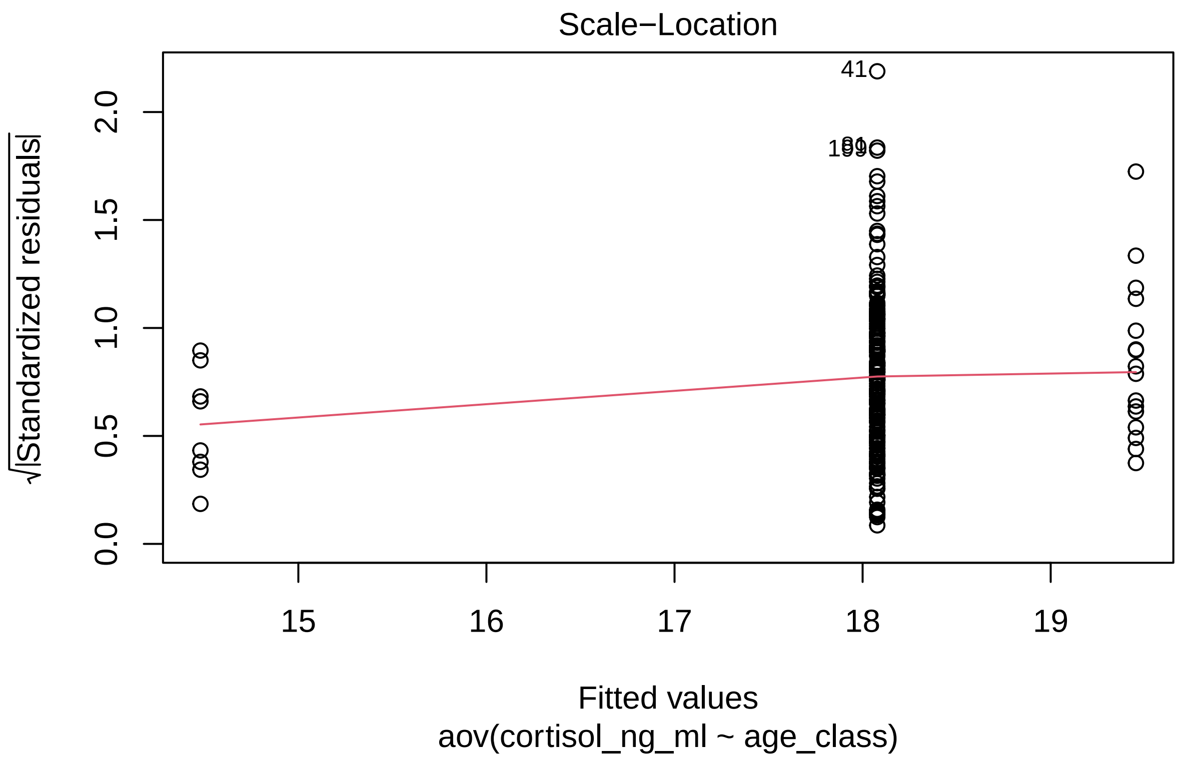
8.3 Model Diagnostic Plots

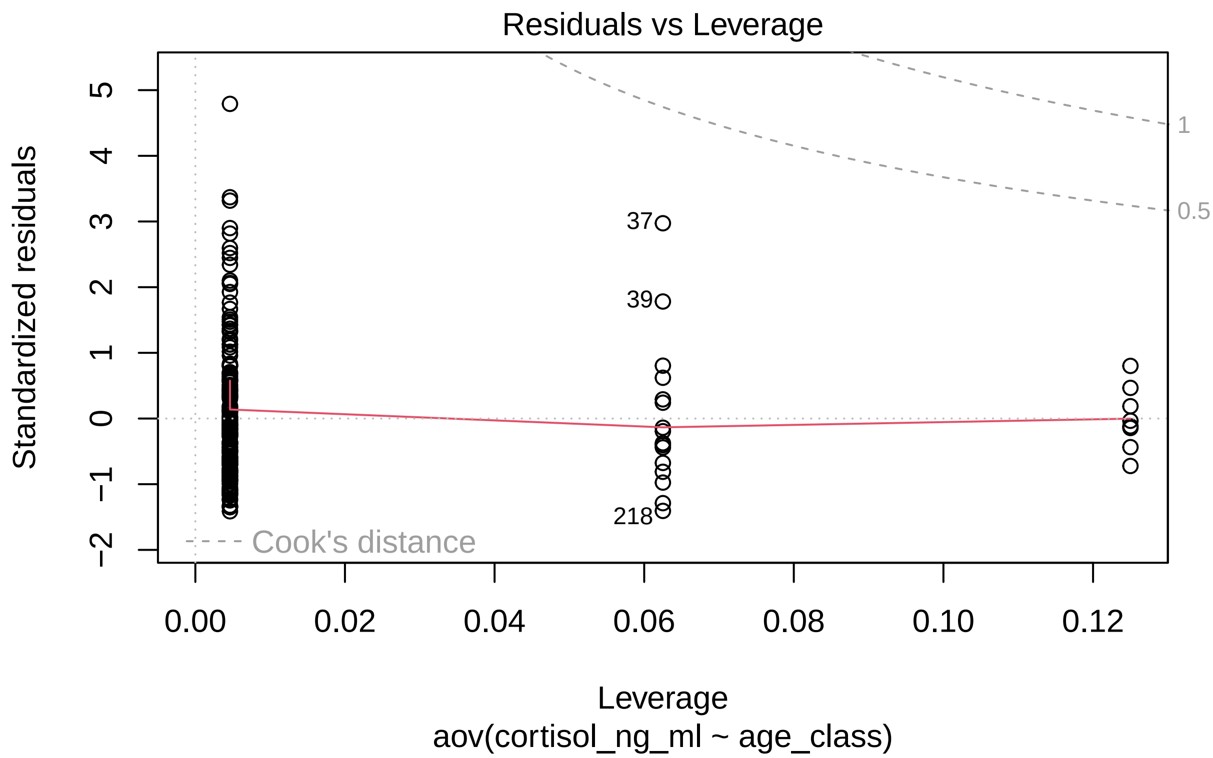
```
mod_aov2 <- aov(cortisol_ng_ml ~ age_class, data = data)

# Diagnostic plots for ANOVA model
plot(mod_aov2)
```


Residuals vs Fitted
41
91
199
Residuals
40
30
20
10
0
−10
15
16
17
18
19
Fitted values
aov(cortisol_ng_ml ~ age_class)

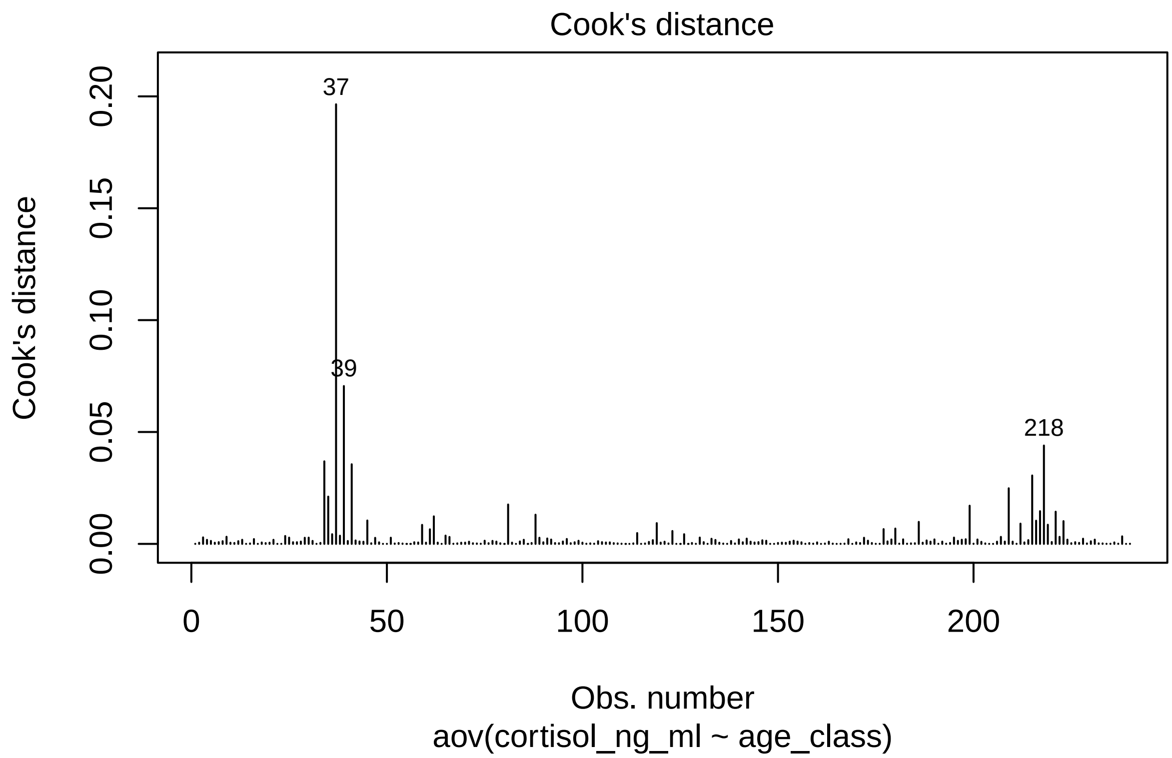


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
aov(cortisol_ng_ml ~ age_class)
41
199
81


Scale−Location
41
191
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
15
16
17
18
19
Fitted values
aov(cortisol_ng_ml ~ age_class)




```
plot(mod_aov2, which = 4)
```




```
cd <- cooks.distance(mod_aov2)

sort(cd, decreasing = TRUE)
```

```
##           37           39          218           34
## 1.964653e-01 7.054055e-02 4.395872e-02 3.688795e-02
##           41          215          209           35
## 3.557982e-02 3.057158e-02 2.484822e-02 2.112354e-02
##           81          199          217          221
## 1.759920e-02 1.706180e-02 1.462128e-02 1.437832e-02
##           88           62           45          216
## 1.303884e-02 1.229320e-02 1.044822e-02 1.035529e-02
##          223          186          119          212
## 1.016106e-02 9.834880e-03 9.277737e-03 9.043954e-03
##          219           59          180          177
## 8.557430e-03 8.501027e-03 6.841244e-03 6.599217e-03
##           61          123          114          126
## 6.509204e-03 5.751175e-03 4.829486e-03 4.315996e-03
##           36           65           38           24
## 4.314520e-03 3.699290e-03 3.647680e-03 3.519819e-03
##          238            9          222          207
## 3.364628e-03 3.175052e-03 3.140146e-03 3.123604e-03
##           66            3          130          195
## 3.092804e-03 2.898864e-03 2.825691e-03 2.820734e-03
##           25           30           51           29
```

```
## 2.799545e-03 2.796823e-03 2.753529e-03 2.736301e-03
##           89          172           47           91
## 2.734256e-03 2.729771e-03 2.694052e-03 2.408910e-03
##          142          228          133           96
## 2.356379e-03 2.309448e-03 2.282792e-03 2.187616e-03
##           16          168          198          182
## 2.139576e-03 2.080133e-03 2.068257e-03 2.026271e-03
##          190          140          179          201
## 2.000871e-03 1.982709e-03 1.978161e-03 1.964283e-03
##           92          231          224          197
## 1.932488e-03 1.905367e-03 1.887748e-03 1.849267e-03
##           85           13           21            4
## 1.830898e-03 1.826088e-03 1.820620e-03 1.789955e-03
##          134          118          214          146
## 1.743576e-03 1.670262e-03 1.669550e-03 1.606065e-03
##           42          188          173          154
## 1.536964e-03 1.532941e-03 1.425559e-03 1.423050e-03
##           75           99          147          196
## 1.417145e-03 1.395418e-03 1.393084e-03 1.356332e-03
##           77            5           31           40
## 1.330530e-03 1.327539e-03 1.326264e-03 1.288175e-03
##          138           12            8          104
## 1.276856e-03 1.213067e-03 1.206047e-03 1.184321e-03
##          230          178           84           43
## 1.165708e-03 1.109342e-03 1.080296e-03 1.070480e-03
##           95           28          155           78
## 1.064505e-03 1.033172e-03 1.024636e-03 1.021171e-03
##          208          192          189          210
## 1.015322e-03 1.005641e-03 9.981779e-04 9.914793e-04
##          153          206          143           71
## 9.877806e-04 9.872882e-04 9.858400e-04 9.760239e-04
##          163          121           44          202
## 9.731807e-04 9.720152e-04 9.634087e-04 9.078590e-04
##          117          220          145          131
## 8.821657e-04 8.291952e-04 7.850136e-04 7.737858e-04
##          105            7          226           27
## 7.677358e-04 7.655023e-04 7.617043e-04 7.493538e-04
##           26           57           98          141
## 7.474069e-04 7.326870e-04 7.283626e-04 7.186776e-04
##           90          156           48          107
## 7.146720e-04 7.124763e-04 6.917131e-04 6.696325e-04
##          160          213          170          236
## 6.675824e-04 6.633425e-04 6.561512e-04 6.492614e-04
##           18           58          106          144
## 6.294740e-04 6.216650e-04 6.212331e-04 6.166543e-04
##          120           63           82           70
## 5.683593e-04 5.655748e-04 5.547958e-04 5.529217e-04
##          151          100           60           20
## 5.523894e-04 5.472044e-04 5.457369e-04 5.286162e-04
##          187          135           69            2
## 5.235955e-04 5.229513e-04 5.179806e-04 5.154553e-04
##            6           10          152          227
## 5.112134e-04 5.047784e-04 4.703964e-04 4.429500e-04
##           33          150           11          128
```

```
## 4.365870e-04 4.142846e-04 4.039790e-04 3.981207e-04
##          194          174           87           53
## 3.960158e-04 3.859715e-04 3.855424e-04 3.800141e-04
##           93          171          158           19
## 3.677039e-04 3.650269e-04 3.539057e-04 3.537946e-04
##          108          225           97           76
## 3.351030e-04 3.112669e-04 3.051453e-04 2.999091e-04
##           94           72          185           79
## 2.983279e-04 2.807048e-04 2.793460e-04 2.792410e-04
##           46          102          184          232
## 2.706802e-04 2.392657e-04 2.368018e-04 2.314342e-04
##           68          109          113          233
## 2.291067e-04 2.218532e-04 2.193626e-04 2.129779e-04
##          203           15          139           73
## 2.089239e-04 2.023232e-04 1.998951e-04 1.972008e-04
##          229          122          116           52
## 1.968812e-04 1.940022e-04 1.819437e-04 1.775252e-04
##          162           54          136          164
## 1.656150e-04 1.588196e-04 1.548913e-04 1.342976e-04
##          181          159          167          110
## 1.197220e-04 1.180432e-04 1.124169e-04 9.956320e-05
##          103          183          200           74
## 9.947746e-05 9.937930e-05 9.692541e-05 9.302777e-05
##          237          193          101          235
## 8.962690e-05 8.419450e-05 7.812601e-05 7.730369e-05
##           67          240          234          166
## 7.284748e-05 6.722741e-05 5.976040e-05 5.941138e-05
##          211          176          129          191
## 5.613464e-05 5.157579e-05 4.761212e-05 4.527108e-05
##          137          157          175           50
## 4.425288e-05 4.213033e-05 4.130800e-05 3.691467e-05
##           23          115            1          132
## 3.308633e-05 3.108658e-05 2.936167e-05 2.632357e-05
##           14          148          204           49
## 2.399210e-05 2.390267e-05 2.309307e-05 2.272934e-05
##           17           64          149          165
## 1.815485e-05 1.688590e-05 1.660066e-05 1.615706e-05
##           32          124          161          169
## 1.582524e-05 1.531778e-05 1.300927e-05 9.662486e-06
##          239          112           83          205
## 7.791044e-06 6.755351e-06 6.581429e-06 3.466653e-06
##           22           86           80          127
## 2.117189e-06 9.719893e-07 8.520673e-07 6.971321e-07
##           55           56          125          111
## 5.778277e-07 4.870803e-07 3.671650e-07 8.416332e-08
```

```r
which(cd > 4 / length(cd))
```

```
##  34  35  37  39  41  81 199 209 215 218
##  34  35  37  39  41  81 199 209 215 218
```

```r
data[which(cd > 4 / nrow(data)), ]
```

```
## # A tibble: 10 x 20
```

```
##    sample_id gorilla_id gorilla_name group  sample_date
##    <chr>     <chr>      <chr>        <chr>  <chr>
##  1 G005_S02  G005       Ubusugire    PAB    3/10/25
##  2 G005_S03  G005       Ubusugire    PAB    12/10/25
##  3 G005_S05  G005       Ubusugire    PAB    3/17/25
##  4 G005_S07  G005       Ubusugire    PAB    1/14/25
##  5 G006_S01  G006       Mushya       ISA    4/1/25
##  6 G011_S01  G011       Nyirindekwe  MSK    7/21/25
##  7 G025_S07  G025       Agahinda     PAB    7/25/25
##  8 G027_S01  G027       Frito        Group4 3/19/25
##  9 G027_S07  G027       Frito        Group4 2/18/25
## 10 G028_S02  G028       Isaro        BEE    2/12/25
## # i 15 more variables: observer <chr>, sex <chr>,
## #   age_class <fct>, age_years <dbl>, dominance_rank <dbl>,
## #   group_size <dbl>, fruit_availability_index <dbl>,
## #   feeding_time_prop <dbl>, aggression_events <dbl>,
## #   parasite_load <dbl>, respiratory_signs <dbl>,
## #   injury_status <dbl>, cortisol_ng_ml <dbl>,
## #   testosterone_ng_ml <dbl>, injury_label <chr>
```

```r
data_cook <- data %>%
  mutate(cooks_d = cooks.distance(mod_aov2)) %>%
  arrange(desc(cooks_d))

# to print them all at once:
# par(mfrow = c(2, 2), oma = c(0, 0, 2, 0))
# plot(mod_aov2, which = 1:4)
```

Residuals vs Fitted: This is the most important diagnostic plot.
Ask: is there any obvious pattern?

Ideally we want:

- random cloud
- no curve
- roughly equal spread

What do we see?

The residuals appear scattered around zero without a strong trend. That's good!

The vertical "stripes" that we see occur because ANOVA predicts one mean for each group, so all observations within a group share the same fitted value. That's exactly what we expect in an ANOVA.

This plot doesn't suggest any obvious problems with the model.

The other two plots are less important right now.

1. Scale-location: is variability similar across groups. Ideally, there a similar spread and a roughly horizontal line. Our plot is acceptable.
2. Cook's distance: identifies observations that have an unudsually large influence on the fitted model.

We have a few that stand out in Cook's distance (e.g., observation 223). It doesn't mean they're wrong, just that if we removed these observations, the model might change more than if we removed most other observations. They deserve a closer look.

A large Cook's distance is a reason to investigate, not to delete. For example, we might ask:

- Was there a data entry error?
- Was the measurement recorded correctly?
- Is this biologically plausible?
- Is it simply an unusually high parasite load?

If the observation is real, it usually stays in the analysis.

9. Summary: Choosing the Right Group Comparison

When comparing groups with a quantitative response variable, ask:

1. What is the response variable?
2. Is the response variable quantitative?
3. What is the grouping variable?
4. How many groups are there?
5. Are the groups independent or paired?
6. Do the data look approximately normal?
7. Which test matches the question and data structure?

Skills Application – Lab

10. Lab Challenge: Compare Groups in Your Own Dataset

For this lab, you will work with your own dataset.

Your goal is to choose a quantitative response variable and compare it across groups.

You do not need to run every test. Focus on choosing the correct test for your question.

Task List

1. Load your dataset and continue working in your script
2. Choose one quantitative response variable
3. Choose one grouping variable; Count how many groups are in your grouping variable
4. Create a plot comparing the response variable across groups
5. Suggest a biological question you can answer using these variables
6. Decide which test is most appropriate and run a test
7. Write 2-3 sentences interpreting the results, biologically

   Goal: perform an analysis from start to finish! Research question –> Data –> Analysis –> Biological Insight